



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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29-04-2025



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
 - Data Collection using Space X API and Web Scraping
 - Data Wrangling
 - Exploratory Data Analysis (EDA) using SQL
 - EDA Data Visualization
 - Interactive Visual Analytics with Folium and Plotly Dash
 - Machine Learning models applied to Space X Landing Prediction
- Summary of all results
 - EDA results
 - Interactive Visual Analytics and Dashboards
 - Predictive Analysis

Introduction



- Project background and context

Space X advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars, other providers cost upward of 165 million dollars each, much of the savings is because Space X can reuse the first stage. Therefore, if we can determine if the first stage will land, we can determine the cost of a launch. This information can be used if an alternate company wants to bid against Space X for a rocket launch.

- Problems you want to find answers

In this work, we will predict the success of the Falcon 9 first stage landing using data from its API and website.

Section 1

Methodology

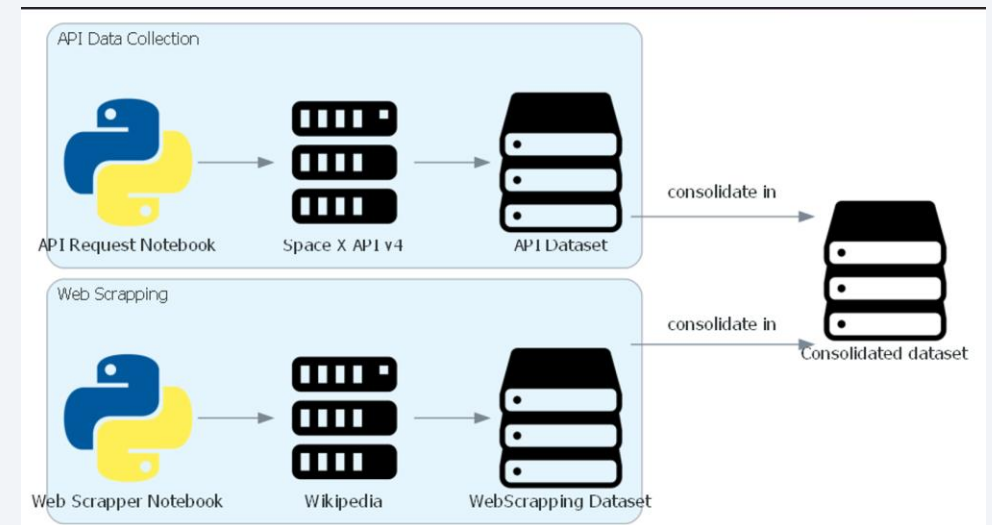
Methodology

Executive Summary

- Data collection methodology:
 - Wikipedia web scraping, Space X API requests
- Perform data wrangling
 - Cleaning empty target values, handling missing values, adjusting data types
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Using GridSearchCV for cross validation and finding the best hyper parameters for Logistic Regression, Support Vector Machine, Decision Tree and KNN models
 - Comparing the performance of each best model

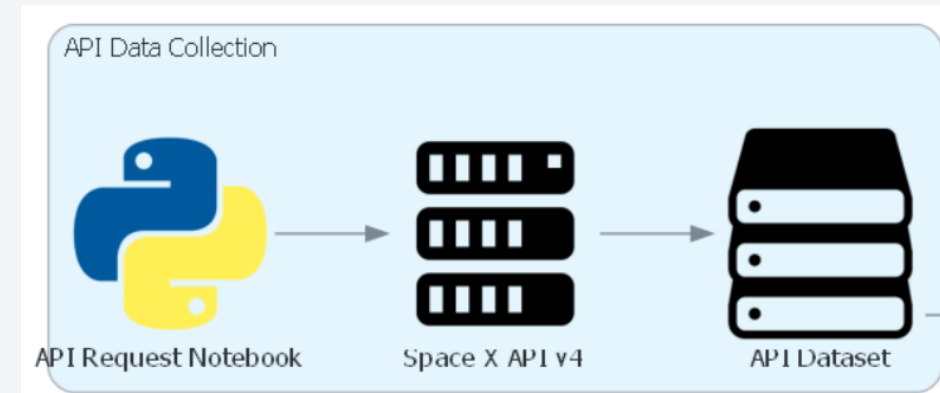
Data Collection

- The data sets were collected by web scraping and using the Space X API
 - The version 4 and the resources rockets, launchpads, payloads and cores of the API were used to generate a consolidated dataset.
 - <https://api.spacexdata.com/v4/>
 - The Wikipedia page titled `List of Falcon 9 and Falcon Heavy Launches` was used to collect Falcon 9 historical launch record performing web scraping
 - https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches



Data Collection – SpaceX API

- We use the version 4 and the resources rockets, launchpads, payloads and cores of the Space X API to generate a consolidated data frame.
 - From the **rocket resource API** we would like to learn the booster name
 - From the **payload resource API** we would like to learn the mass of the payload and the orbit that it is going to
 - From the **launchpad resource API** we would like to know the name of the launch site being used, the longitude, and the latitude.
 - From the **cores resource API** we would like to learn the outcome of the landing, the type of the landing, number of flights with that core, whether grid fins were used, whether the core is reused, whether legs were used, the landing pad used, the block of the core which is a number used to separate version of cores, the number of times this specific core has been reused, and the serial of the core.
- GitHub repo (https://github.com/MeloDev2111/IBM-Data-Science-Learning-Path---Notebook/blob/main/Course%2010%20-%20Applied%20DS%20Capstone/Lab1_Collecting_Data.ipynb)

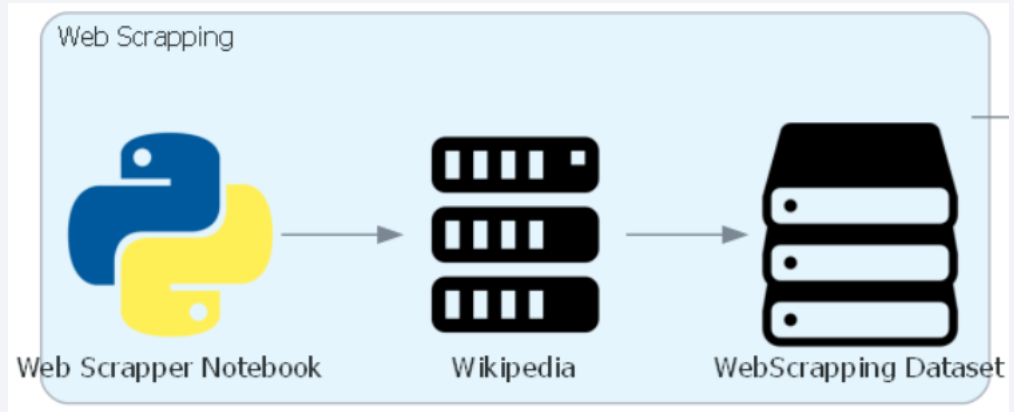


```
launch_df = pd.DataFrame(launch_dict)
launch_df.head(2)
```

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Le
0	1	2006-03-24	Falcon 1	20.0	LEO	Kwajalein Atoll	None None	1	False	False	Fal
1	2	2007-03-21	Falcon 1	NaN	LEO	Kwajalein Atoll	None None	1	False	False	Fal

Data Collection - Scraping

- We perform web scraping on the Wikipedia page titled `List of Falcon 9 and Falcon Heavy Launches` to collect Falcon 9 historical launch record performing web scraping
- Github Repo
(https://github.com/MeloDev2111/IBM-Data-Science-Learning-Path---Notebook/blob/main/Course%2010%20-%20Applied%20DS%20Capstone/Lab2_Web scraping.ipynb)



```
Flight No. 1
date 4 June 2010
time 18:45
Booster Version F9 v1.07B0003.18
Launch site CCAFS
Launch outcome Success

Booster landing Failure

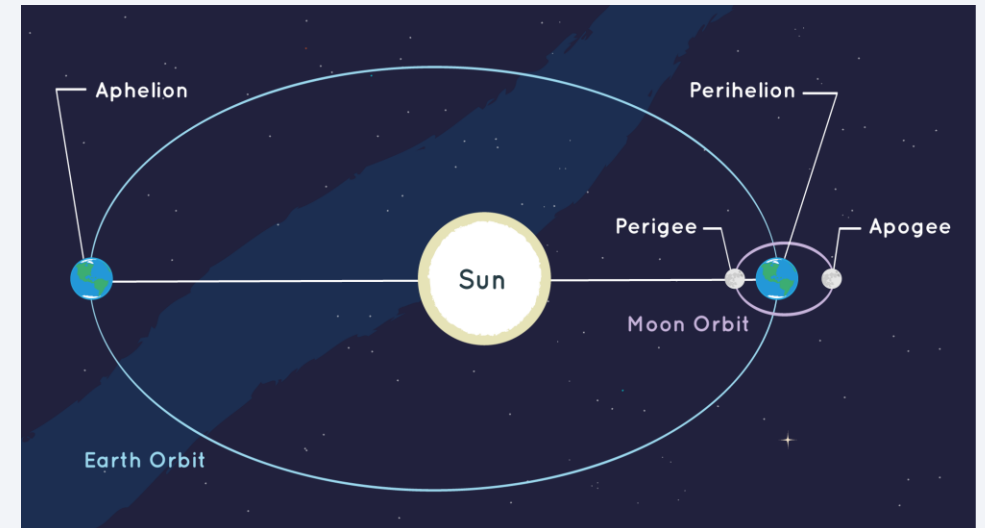
Flight No. 2
date 8 December 2010
time 15:43
Booster Version F9 v1.07B0004.18
Launch site CCAFS
Launch outcome Success
Booster landing Failure

Flight No. 3
date 22 May 2012
time 07:44
Booster Version F9 v1.07B0005.18
Launch site CCAFS
Launch outcome Success
Booster landing No attempt

Extracted 121 rows
```

Data Wrangling

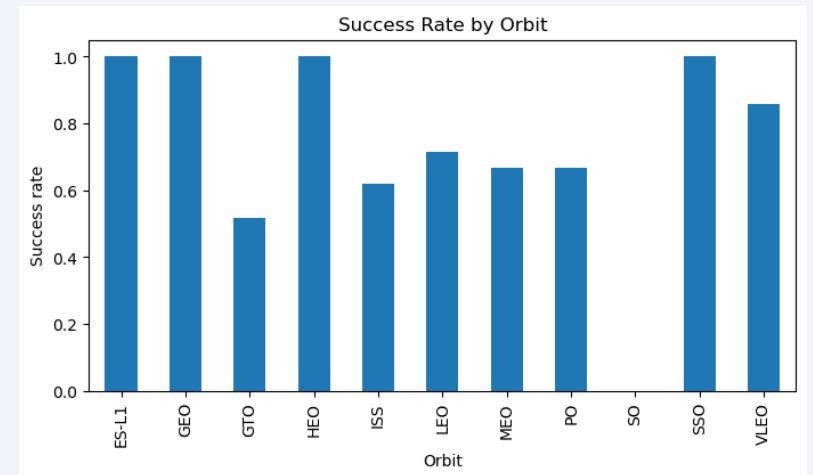
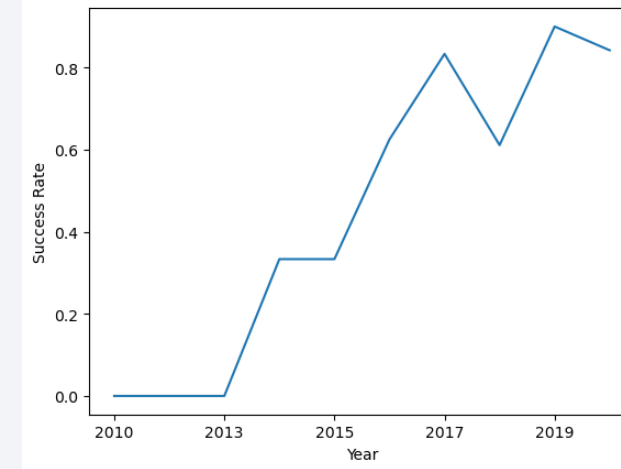
- We determine the training labels after
 - Calculating of the percentage of the missing values in each attribute
 - Identifying columns numerical and categorical
- Calculate key metrics as:
 - Numbers of launches of each site
 - Number and occurrence of each orbits
 - Number and occurrence of mission outcome of the orbits
- Create a lading outcome label from Outcome column
- GitHub repo ([https://github.com/MeloDev2111/IBM-Data-Science-Learning-Path---Notebook/blob/main/Course%2010%20-%20Applied%20DS%20Capstone/Lab3 Data Wrangling.ipynb](https://github.com/MeloDev2111/IBM-Data-Science-Learning-Path---Notebook/blob/main/Course%2010%20-%20Applied%20DS%20Capstone/Lab3%20Data%20Wrangling.ipynb))



EDA with Data Visualization

- Charts Plotted

- Flight Number vs Payload Mass, Flight Number vs Launch Site, Payload Mass vs Launch Site, Flight Number vs Orbit Type, Payload and Orbit Type
 - Type: Scatter Plot - Goal: Understand the correlation between the variables
- Launch Success Yearly Trend (Success Rate vs Date)
 - Type: Line Plot – Goal: Visualize the evolution of the average launch success trend over years.
- Success Rate of each Orbit
 - Type: Bar Plot - Goal: Check if there are any relationship between success rate and orbit type



EDA with SQL

- Q1: Display the names of the unique launch sites in the space mission
- Q2: Display 5 records where launch sites begin with the string 'CCA'
- Q3: Display the total payload mass carried by boosters launched by NASA (CRS)
- Q4: Display average payload mass carried by booster version F9 v1.1
- Q5: List the date when the first successful landing outcome in ground pad was achieved.
- Q6: List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
- Q7: List the total number of successful and failure mission outcomes
- Q8: List all the booster_versions that have carried the maximum payload mass.
- Q9: List the records which will display the month names, failure landing_outcomes in drone ship , booster versions, launch_site for the months in year 2015.
- Q10: Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

Build an Interactive Map with Folium

- Use Folium Marker, Popup, Icon and Circle to visualize each launch site based on its coordinate values.
- Use Folium Marker, Marker Cluster and Icon to visualize the success/failed launches for each site on the map.
 - **Red** markers represents rocket launch failures
 - **Green** markers represents the successes launches
- Use Folium Mouse Position to get coordinate for a mouse over a point on the map and use Folium Polyline to draw the distances between a launch site to its proximities

Build a Dashboard with Plotly Dash

- Built a Pie Chart for the Succeed by launch site and give a dropdown filter with each launch site as options and an overall option
- Build a Scatter Plot to visualize the correlation between Payload and Success Outcome. It can be filtered by the dropdown launch site and a scaler filter for payload
- Also, we add a hue on each plot based on launch site, in the general pie chart and the scatter plot.

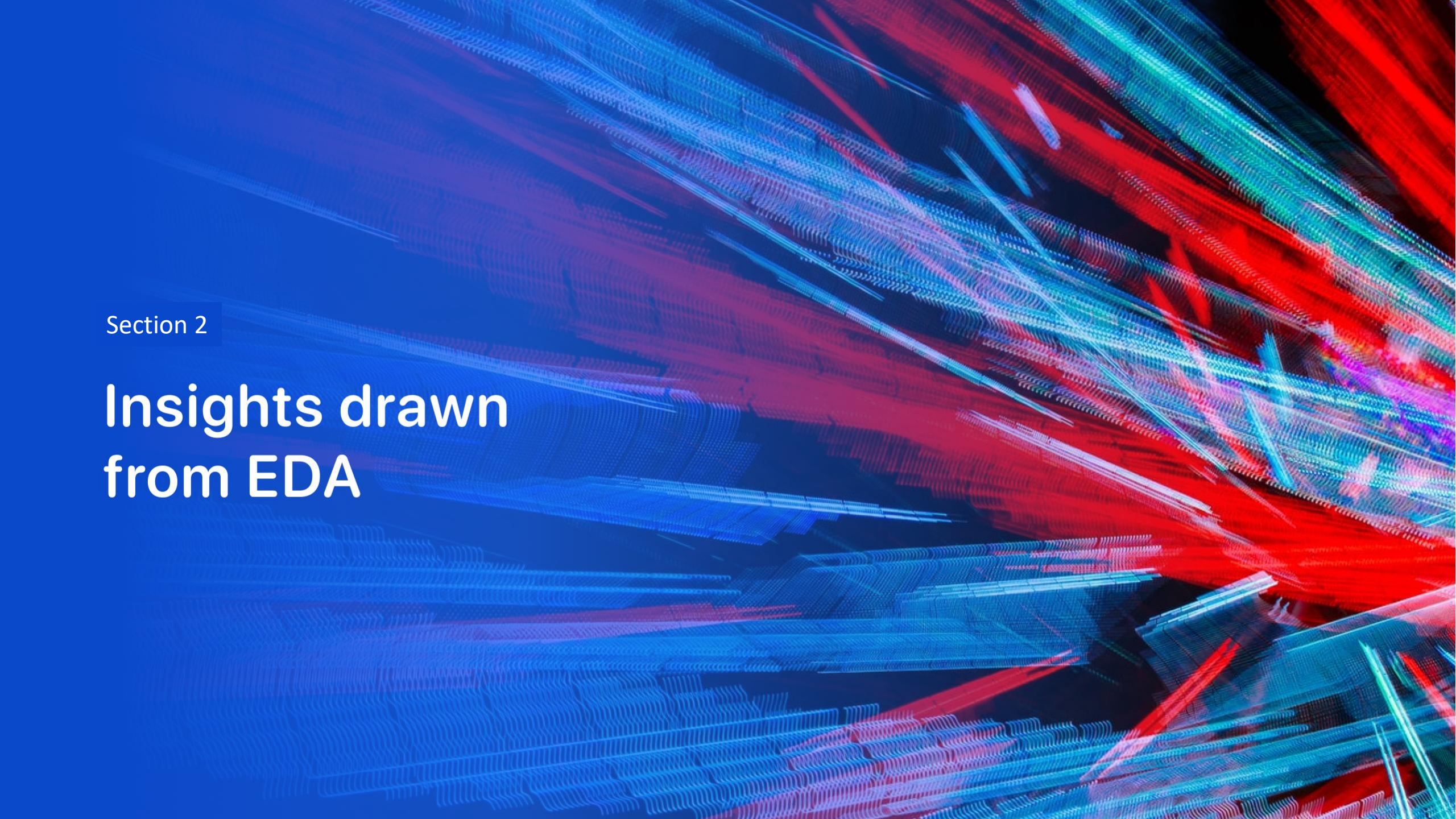
Predictive Analysis (Classification)

- Identify the Target variable
- Standardize the data
- Split the data into test and training with a test size of 0.2
- Use GridSearchCV for cross validation and finding the best hyper parameters for Logistic Regression, Support Vector Machine, Decision Tree and KNN models
- Comparing the performance on training and test data of each model with hyper paremeters tuned

GitHub Repo(https://github.com/MeloDev2111/IBM-Data-Science-Learning-Path---Notebook/blob/main/Course%2010%20-%20Applied%20DS%20Capstone/Lab8_ML_models.ipynb)

Results

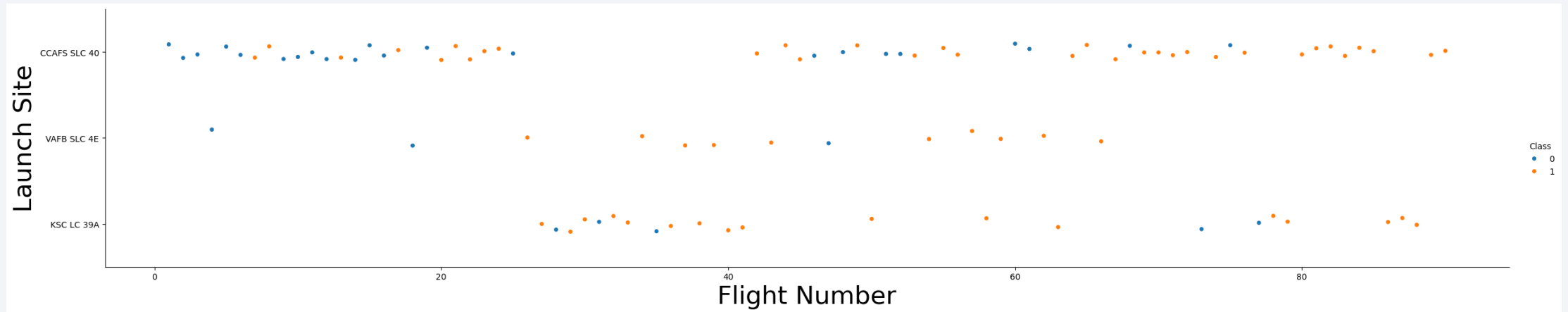
- EDA has show us that the success rate since 2013 kept increasing significantly till 2020
- We observe some of the consideration to place a launch site like the proximity to coast line, equator line, railways, highways and cities. Due to the need of testing rocket landings in the water, facilitate the equipment and research material transportation.
- The tuned machine learnings models were able to predict the landing success of rockets with an accuracy score approximately of 83.333%

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is high-tech and digital.

Section 2

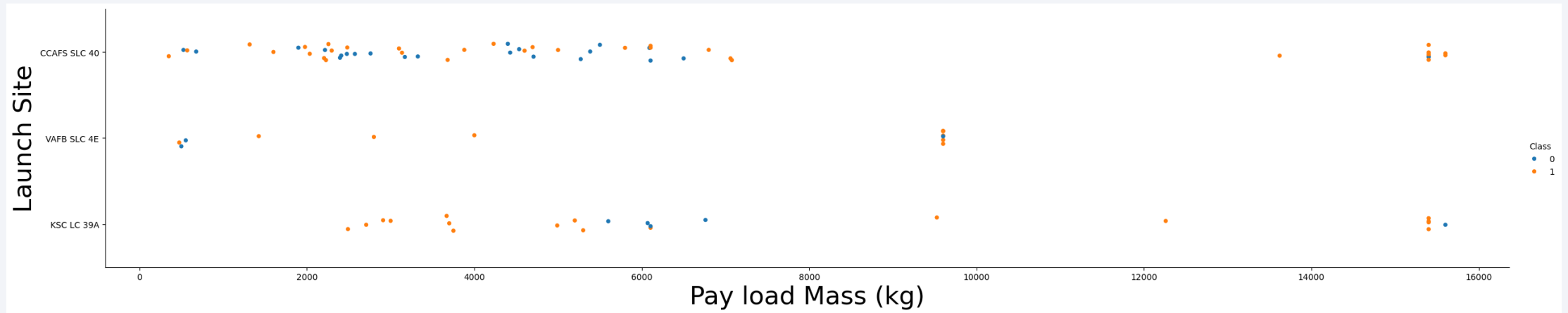
Insights drawn from EDA

Flight Number vs. Launch Site



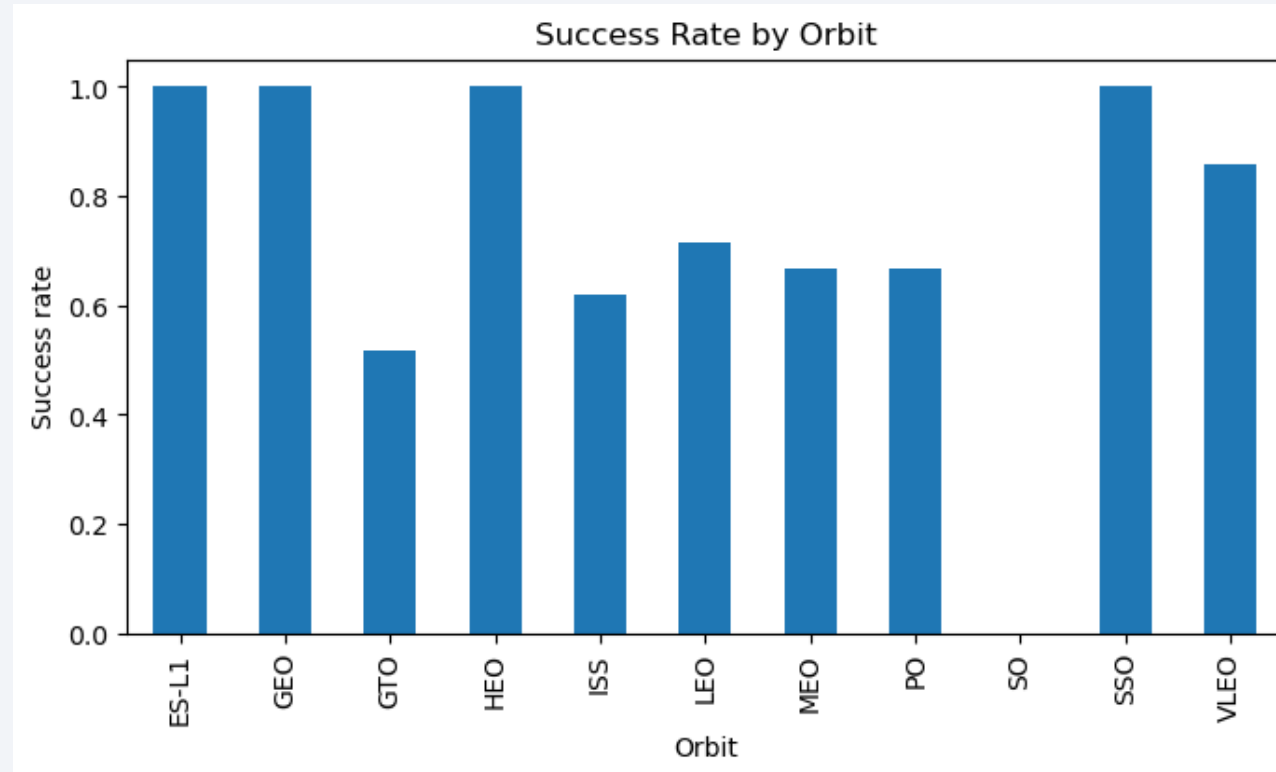
There were more successful landing as the flight numbers increased.

Payload vs. Launch Site



The VAFB-SLC 4E launch site there are no rockets launched for heavy payload mass(greater than 10000).

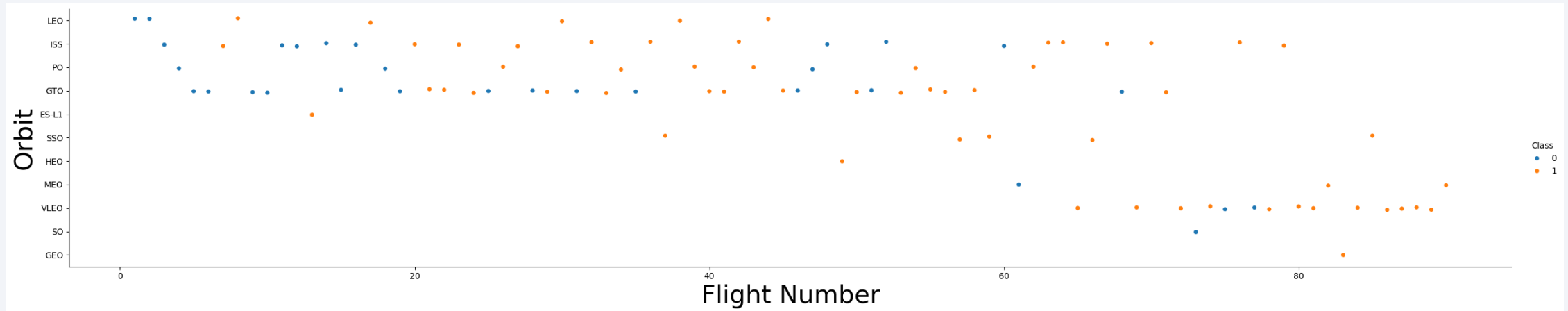
Success Rate vs. Orbit Type



The orbits that have the highest success rates are ES-L1, GEO, HEO and SSO.

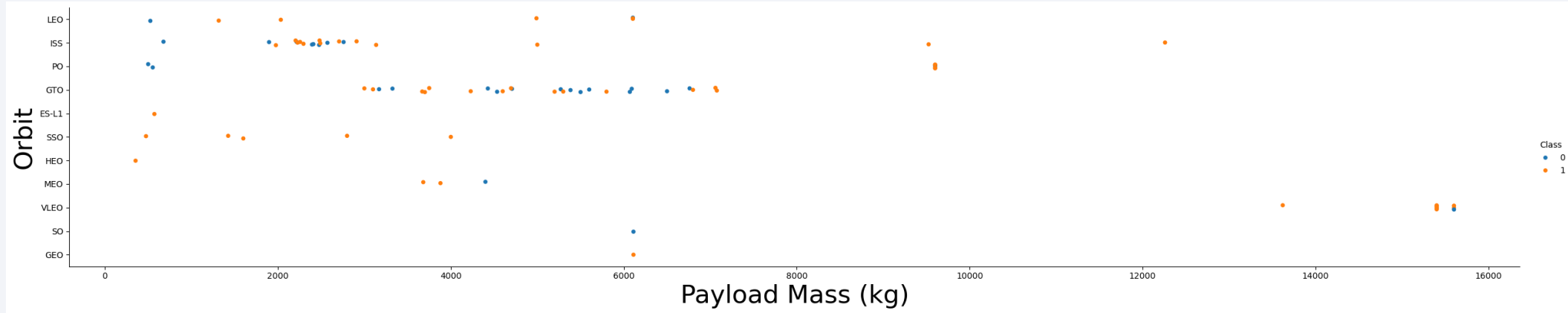
SSO is the one with more flights (5)

Flight Number vs. Orbit Type



You can observe that in the LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.

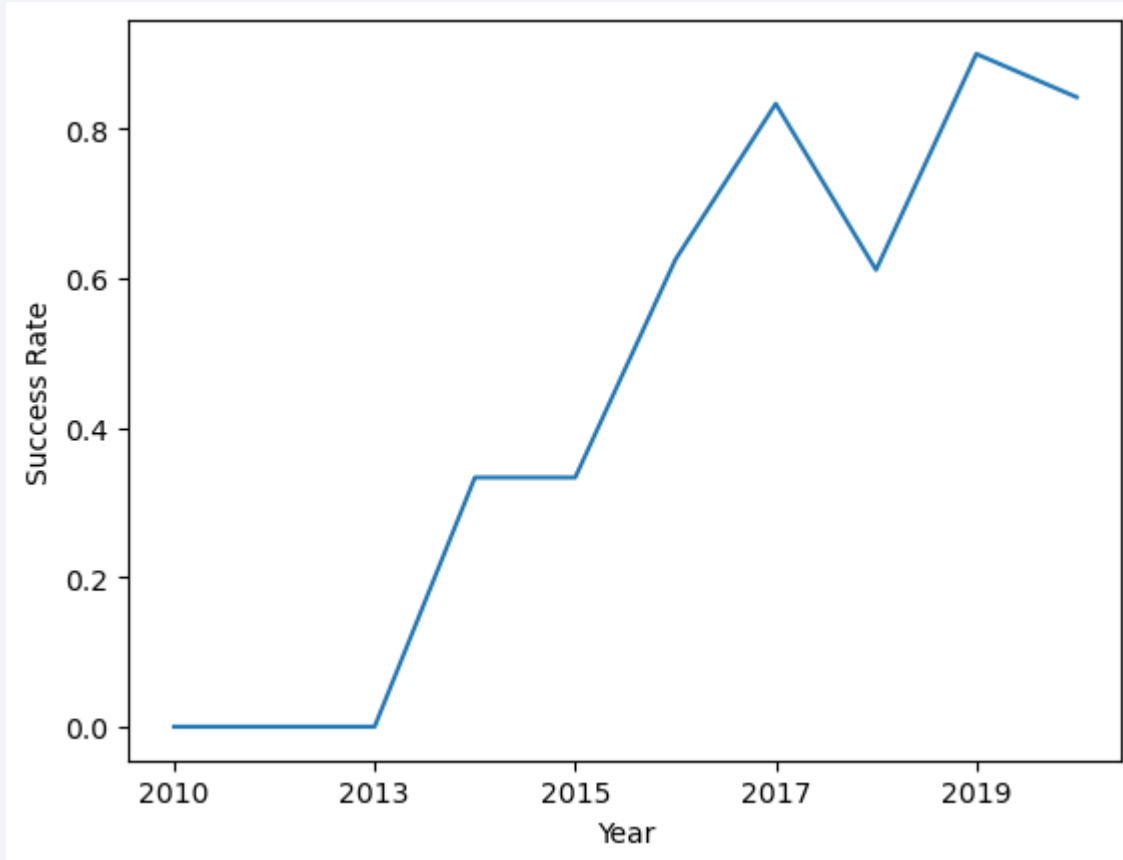
Payload vs. Orbit Type



With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS.

However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.

Launch Success Yearly Trend



You can observe that the success rate since 2013 kept increasing significantly till 2020

All Launch Site Names

The names of the unique launch sites in the space mission are:

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- Some records where launch sites begin with “CCA” are:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success

Total Payload Mass

- The total payload mass carried by boosters launched by NASA (CRS) is 45 596 Kg

total
45596

Average Payload Mass by F9 v1.1

- The average payload mass carried by booster version F9 v1.1 is 2928.4 Kg

average
2928.4

First Successful Ground Landing Date

- The first successful landing outcome on ground pad was on 22/12/2015

```
min(Date)
```

```
2015-12-22
```

Successful Drone Ship Landing with Payload between 4000 and 6000

- The names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000 are:

Booster_Version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- The total number of successful and failure mission outcomes are 100 successful and 1 failure

Mission_Outcome	count(*)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

The names of the booster which have carried the maximum payload mass are:

- F9 B5 B1048.4
- F9 B5 B1049.4
- F9 B5 B1051.3
- F9 B5 B1056.4
- F9 B5 B1048.5
- F9 B5 B1051.4
- F9 B5 B1049.5
- F9 B5 B1060.2
- F9 B5 B1058.3
- F9 B5 B1051.6
- F9 B5 B1060.3
- F9 B5 B1049.7

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- The failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015 are:

month	Landing_Outcome	Booster_Version	Launch_Site	Date	Time (UTC)	Booster_Version_1	Launch_Site_1	Payload
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40	2015-01-10	9:47:00	F9 v1.1 B1012	CCAFS LC-40	SpaceX CRS-5
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40	2015-04-14	20:10:00	F9 v1.1 B1015	CCAFS LC-40	SpaceX CRS-6

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- The rank based on the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order is:

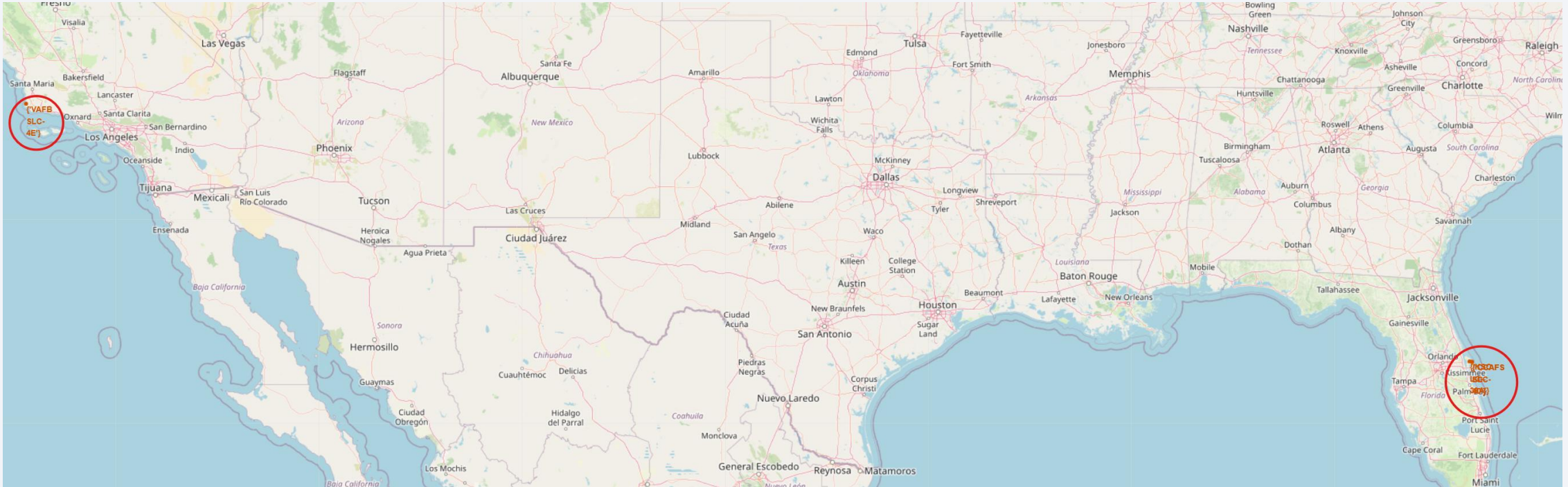
Landing_Outcome	count(*)
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky with stars and a view of the Earth's surface from space. The Earth's surface is mostly dark, with a dense network of yellow and orange lights representing city lights at night. The lights are concentrated in the lower right portion of the image, following the curve of the Earth. The upper portion of the image shows the dark blue sky with a few stars.

Section 3

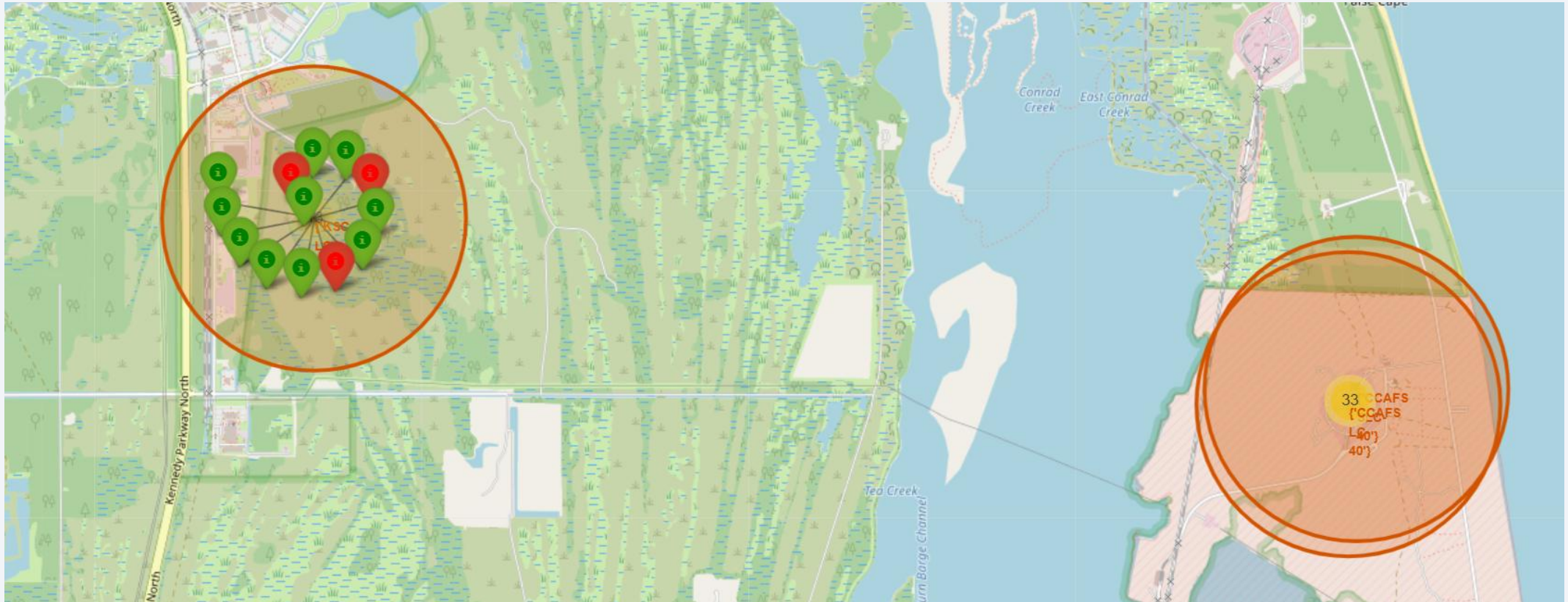
Launch Sites Proximities Analysis

All Launch Sites Locations



- All launch sites are in proximity to the Equator line
- All launch sites are in very close proximity to the coast

Outcome Launch Status by Launch Site

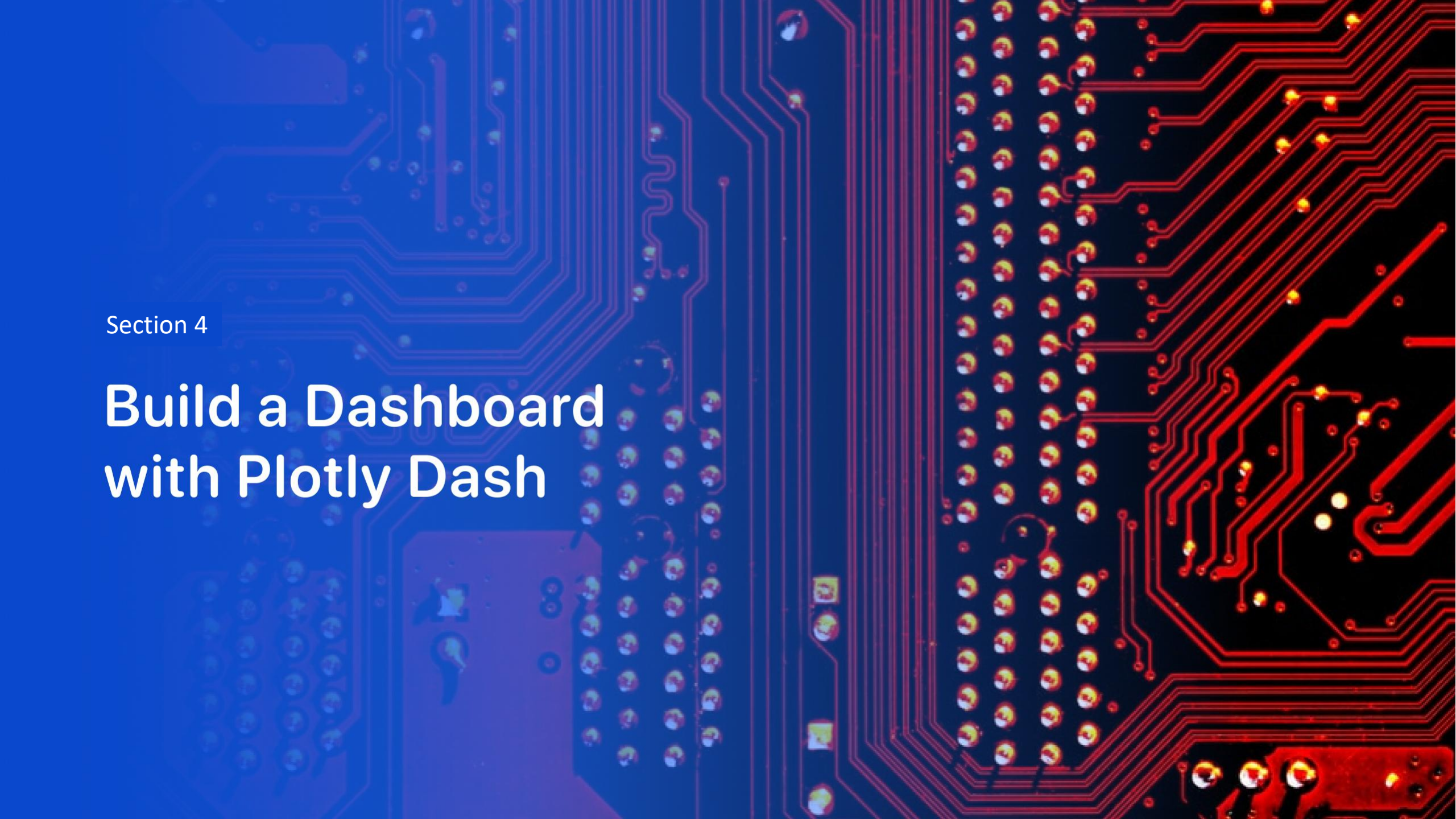


- CCAFS LC-40 is the launch site with more flights (26)
- KSCCLC-39A IS the launch site with more flight successes (10)

Proximities Calculated Distances



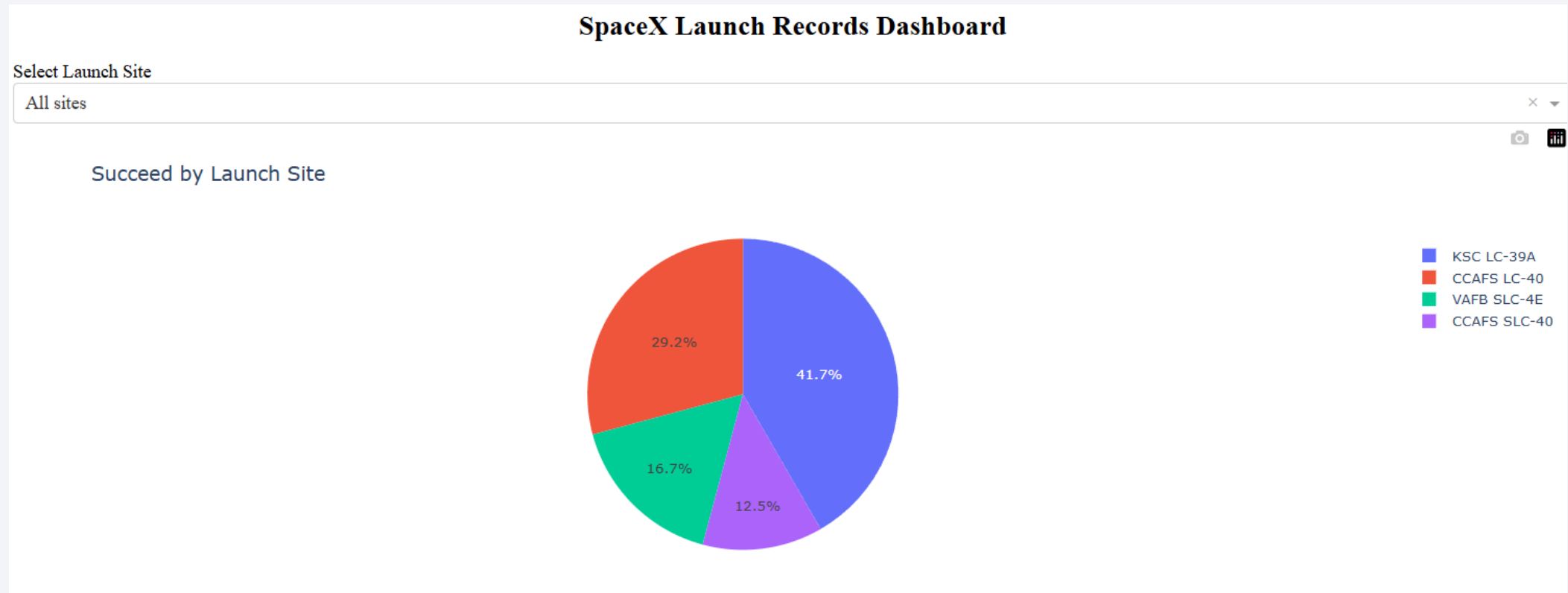
All launch sites are in very close proximity to the coast, railways and highways, but are usually away from cities at least 18 km.



Section 4

Build a Dashboard with Plotly Dash

Success Outcome Count for All Sites



- KSC LC-39A has the highest successful launches count and the highest success rate

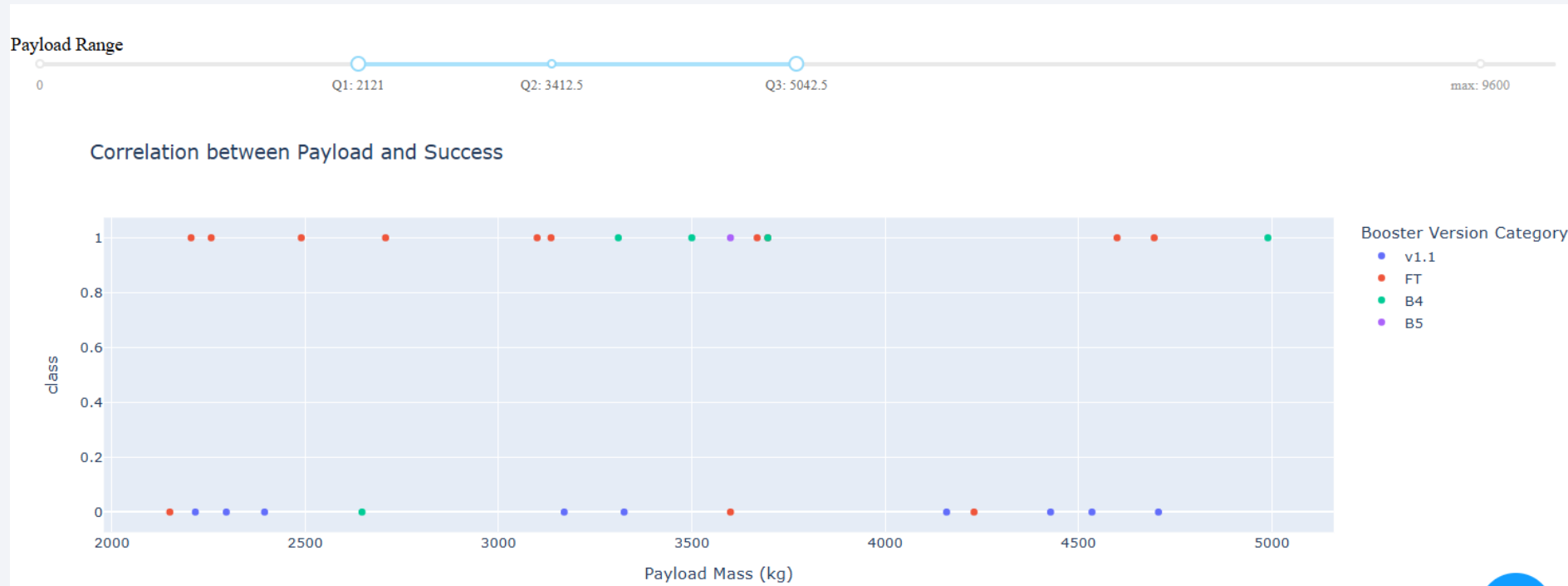
Successful Launches of KSC LC-39A

Succeed by Launch Site



- The success ratio of KSC LC-39A is 0.769. It's the launch site with the highest launch success ratio

Payload Vs Launch Outcome for All Sites



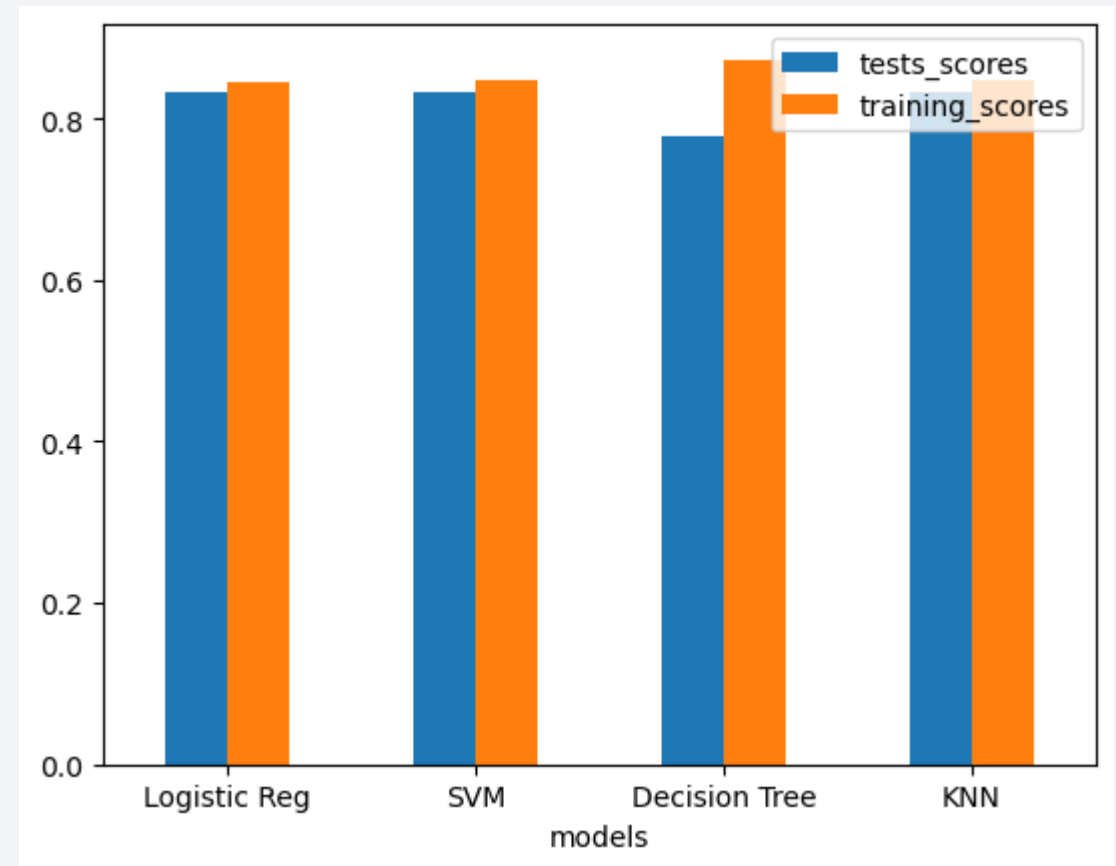
- The payload range between Q1(2121) and Q3(5042) has the highest success rate.

Section 5

Predictive Analysis (Classification)

Classification Accuracy

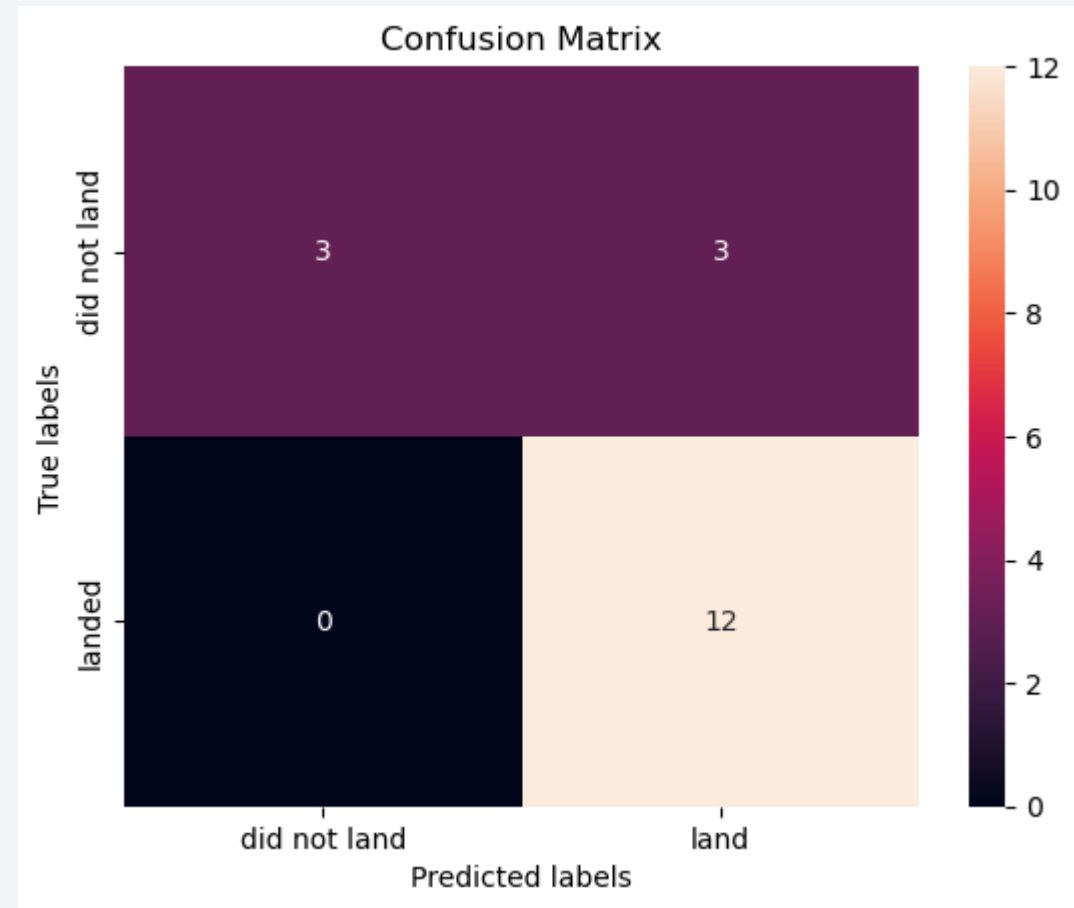
- The model's stats are:
 - **Logistic Regression:** training accuracy 0.846, with test data: 0.8333
 - **SVM:** training accuracy 0.8482142857142856, with test data: 0.8333
 - **Decision Tree:** training accuracy 0.875, with test data: 0.7778
 - **KNN:** training accuracy 0.8482142857142858, with test data: 0.8333



KNN model with tuned hyper parameters: {'algorithm': 'auto', 'n_neighbors': 10, 'p': 1} has the highest accuracy

Confusion Matrix of tuned KNN model

- The model predict 3 false positives with a predicted value of landing successful and true value of “not successful land”. This reduce the accuracy of the model to 83.33%



KNN model with tuned hyper parameters: {'algorithm': 'auto', 'n_neighbors': 10, 'p': 1}

Conclusions

- EDA has show us that the success rate since 2013 kept increasing significantly till 2020
- We observe some of the consideration to place a launch site like the proximity to coast line, equator line, railways, highways and cities. Due to the need of testing rocket landings in the water, facilitate the equipment and research material transportation.
- The tuned machine learnings models were able to predict the landing success of rockets with an accuracy score approximately of 83.333%

Appendix

- Python code snippet used to generate data collection diagram

```
from diagrams import Diagram, Cluster, Edge
from diagrams.programming.language import Python
from diagrams.onprem.compute import Server
from diagrams.generic.storage import Storage

import os
os.environ["PATH"] += os.pathsep + 'C:/Program Files/Graphviz/bin/'

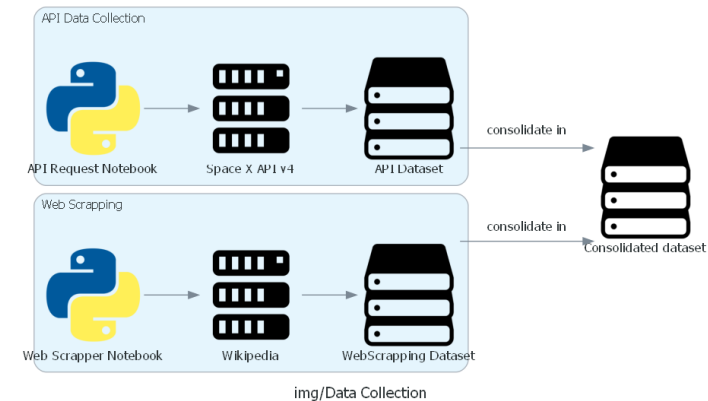
with Diagram("img/Data Collection", show=True):

    d3 = Storage("Consolidated dataset")

    with Cluster("API Data Collection"):
        s1 = Python("API Request Notebook")
        api = Server("Space X API v4")
        d1 = Storage("API Dataset")
        s1 >> api >> d1

    with Cluster("Web Scrapping"):
        s2 = Python("Web Scrapper Notebook")
        wikipedia = Server("Wikipedia")
        d2 = Storage("WebScrapping Dataset")
        s2 >> wikipedia >> d2

    d1 >> Edge(label="consolidate in") >> [d3]
    d2 >> Edge(label="consolidate in") >> [d3]
```



Thank you!

